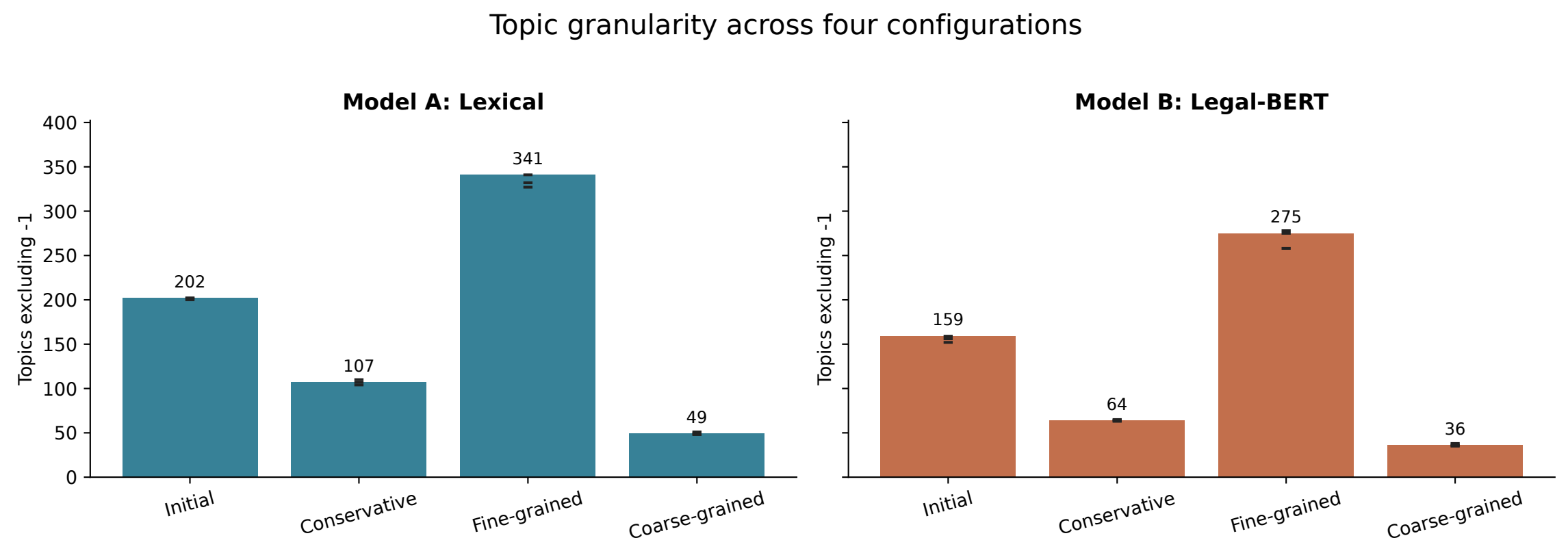
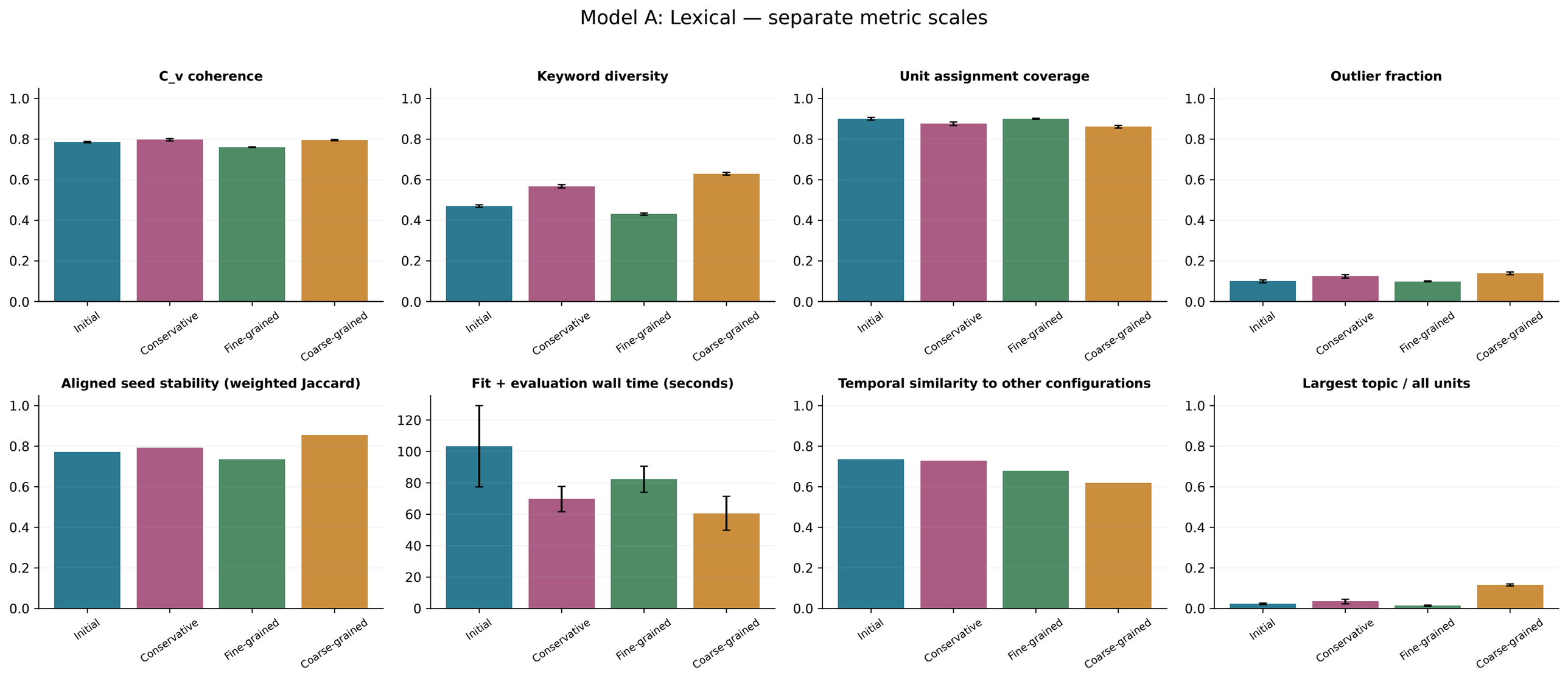
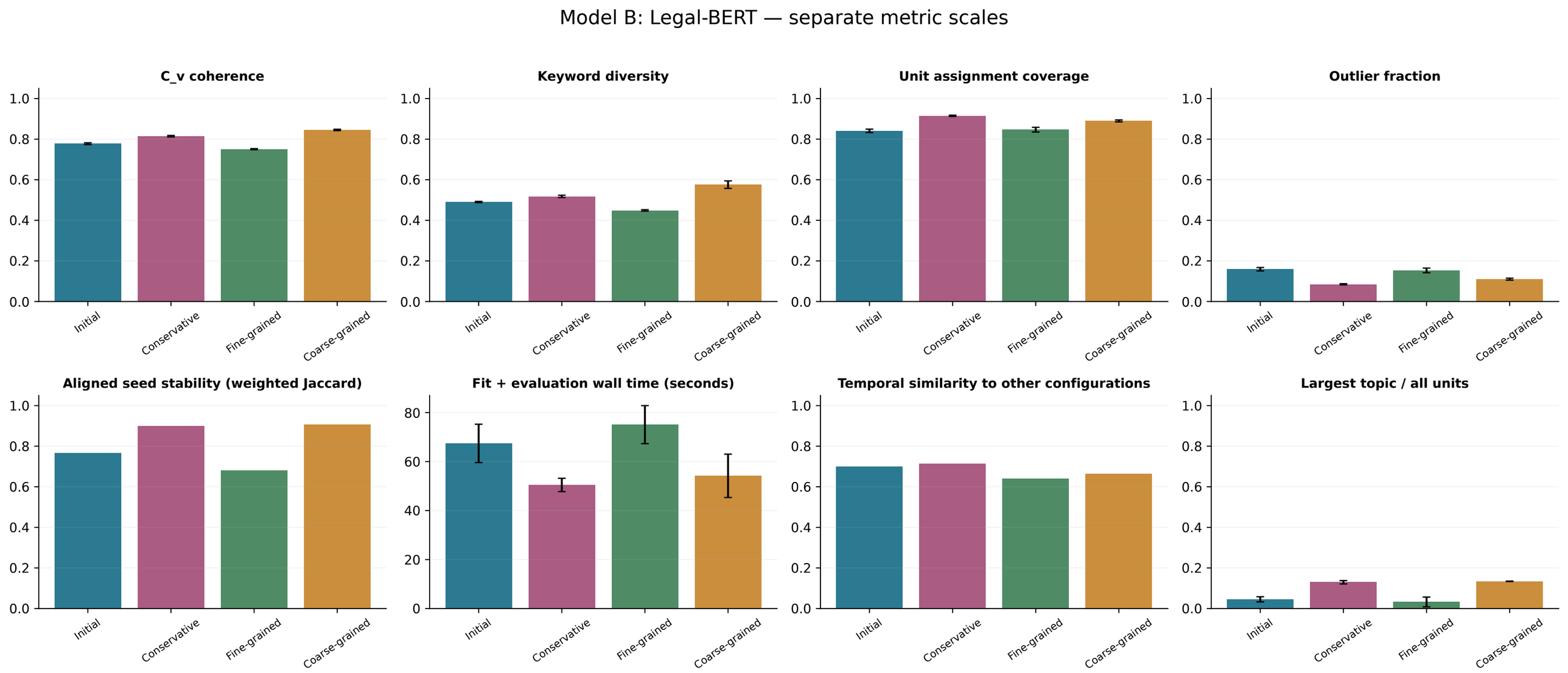


Bars: seed 42; black ticks: individual seeds 42, 43, 44. More topics is not necessarily better.



Means ± sample SD over three seeds where repeated. Stability: three seed pairs. Temporal: seed-42 pairwise comparisons; sensitivity, not truth.
Coverage and outlier fraction are complements. Runtime excludes shared representation construction. No normalized composite quality score.


Model B: Legal-BERT — separate metric scales
C_v coherence
Keyword diversity
Unit assignment coverage
Outlier fraction
Aligned seed stability (weighted Jaccard)
Fit + evaluation wall time (seconds)
Temporal similarity to other configurations
Largest topic / all units
Initial
Conservative
Fine-grained
Coarse-grained
Means ± sample SD over three seeds where repeated. Stability: three seed pairs. Temporal: seed-42 pairwise comparisons; sensitivity, not truth.
Coverage and outlier fraction are complements. Runtime excludes shared representation construction. No normalized composite quality score.

Parameter–performance associations (joint changes, not isolated effects)

Model A: Lexical · Model B: Legal-BERT · Initial · Conservative · Fine-grained · Coarse-grained

C_v (seed mean)

Coverage (seed mean)

Aligned seed stability

Topics (seed mean)

UMAP neighbors

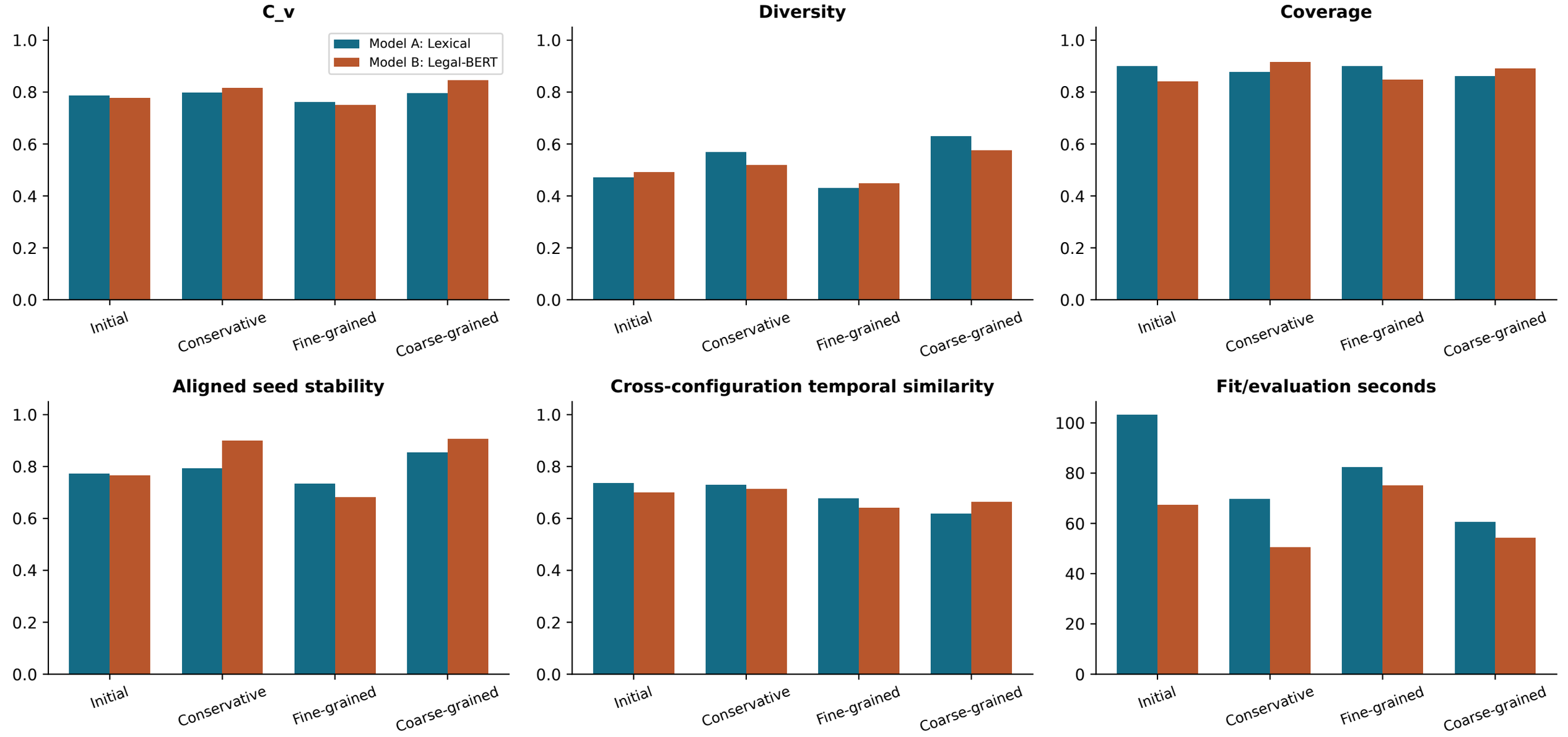
UMAP minimum distance

HDBSCAN minimum cluster size

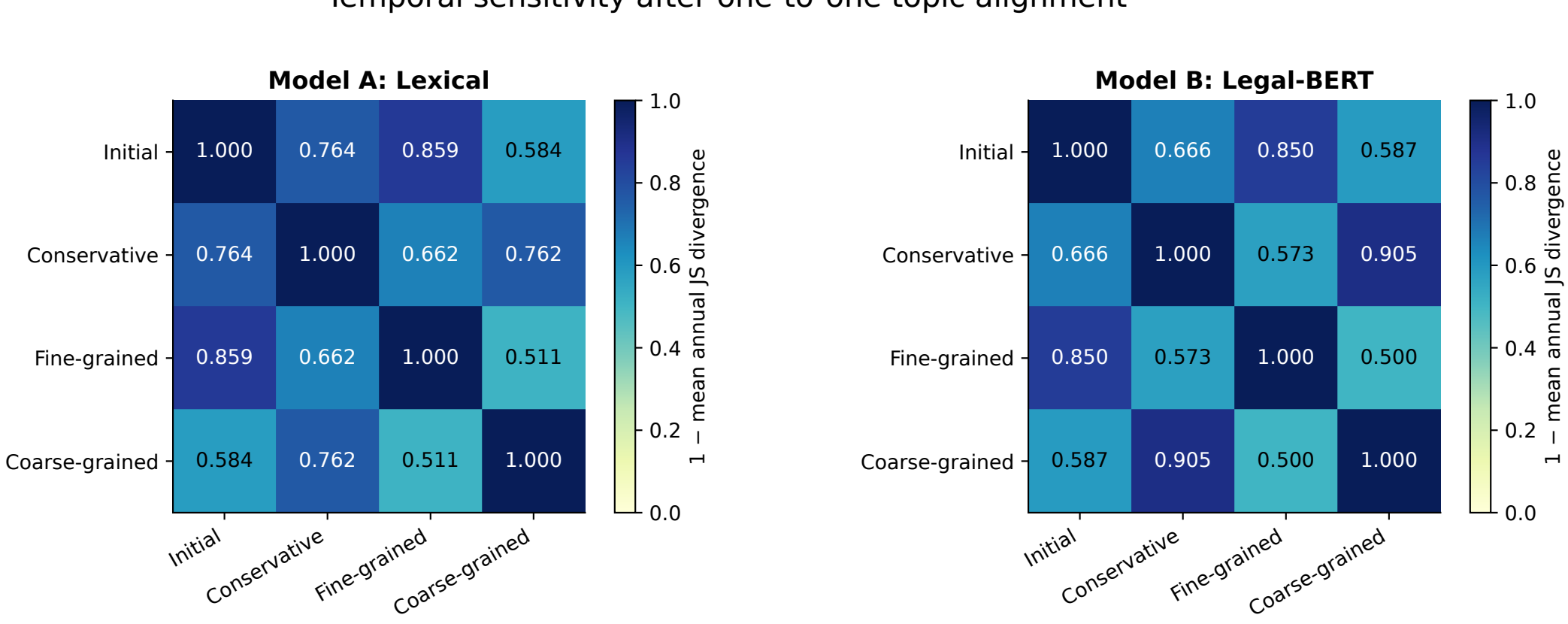
HDBSCAN minimum samples

Four predeclared settings per model. Several parameters co-vary; no regression, causal importance or independent parameter optimum is inferred.
Fine vs Initial fixes UMAP; Coarse vs Conservative fixes UMAP. Marker overlaps are possible.

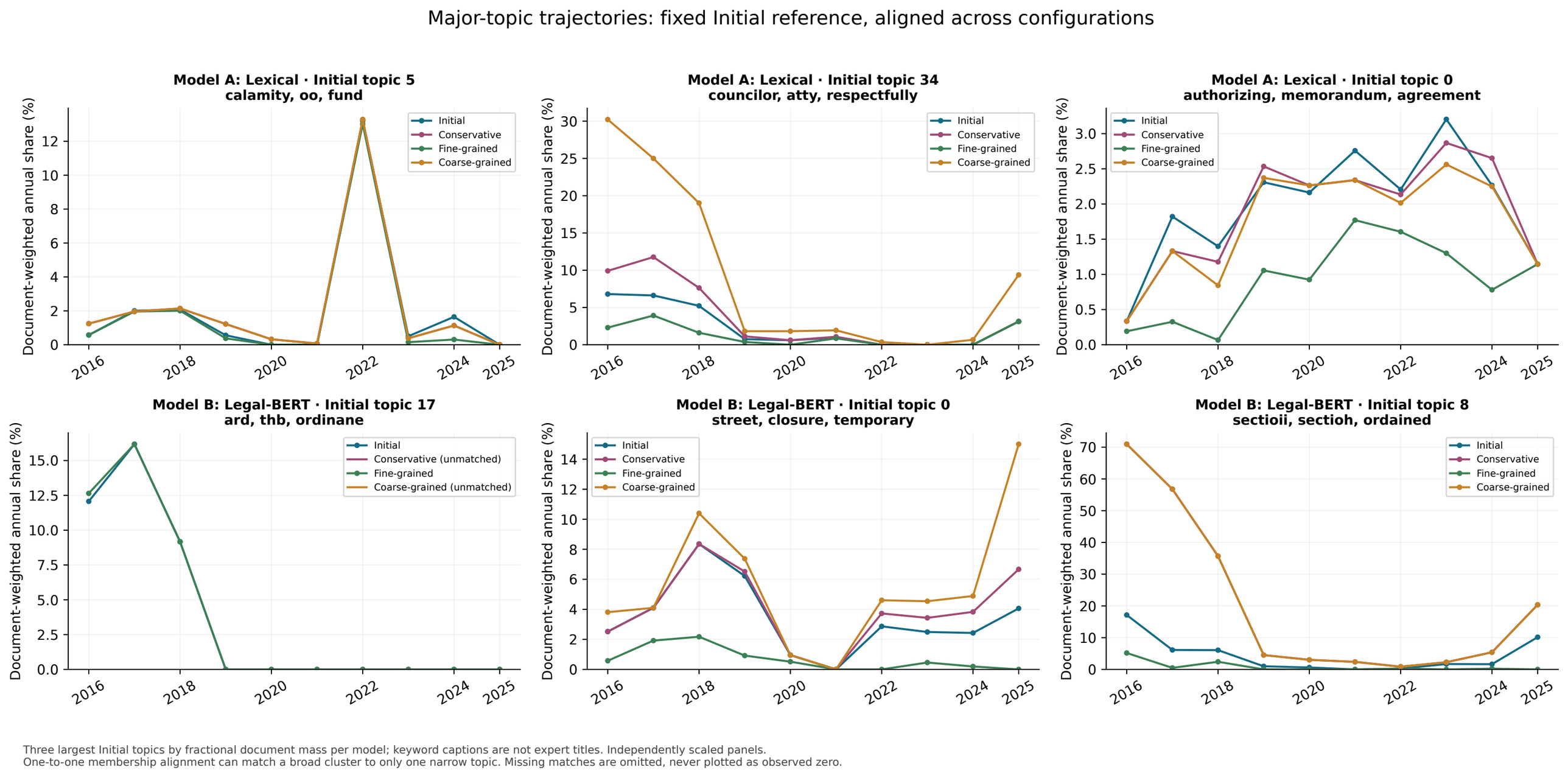
Comparable measurements on identical units and vocabulary



Topic IDs are not shared between models. Cross-configuration sensitivity compares each model with itself, not identical semantic categories.
Seed means except pairwise sensitivity measures. Intrinsic comparisons are not held-out accuracy or expert legal assessment.



Seed 42, document-weighted annual distributions; include unmatched-topic mass and noise. Equal year weights, base-2 divergence.
Matching penalizes topic splits/merges; agreement can include shared outlier mass and does not establish historical validity.


Major-topic trajectories: fixed Initial reference, aligned across configurations
Model A: Lexical · Initial topic 5
calamity, oo, fund
Model A: Lexical · Initial topic 34
councilor, atty, respectfully
Model A: Lexical · Initial topic 0
authorizing, memorandum, agreement
Model B: Legal-BERT · Initial topic 17
ard, thb, ordinane
Model B: Legal-BERT · Initial topic 0
street, closure, temporary
Model B: Legal-BERT · Initial topic 8
sectioii, sectioh, ordained
Document-weighted annual share (%)
Initial
Conservative
Conservative (unmatched)
Fine-grained
Coarse-grained
Coarse-grained (unmatched)
2016
2018
2020
2022
2024
2025
Three largest Initial topics by fractional document mass per model; keyword captions are not expert titles. Independently scaled panels.
One-to-one membership alignment can match a broad cluster to only one narrow topic. Missing matches are omitted, never plotted as observed zero.